MILITARY SPECIFICATION

ISOPROPYLAMINE, TECHNICAL

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of isopropylamine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR 171 to 179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D 891 - Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
D 1078 - Distillation Range of Volatile Organic Liquids
E 203 - Water Using Karl Fischer Reagent

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. Isopropylamine shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Minimum	Maximum	Test paragraph
Assay as isopropylamine, percent by weight	99.0	----	4.2.4.1
Water content, percent by weight	----	0.6	4.2.4.2
Specific gravity at 20°/20°C	0.686	0.690	4.2.4.3
Percent by volume distilled between 30.5° and 34.5°C at 760 mm mercury	90	----	4.2.4.4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the isopropylamine produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling for examination of packaging shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. See 6.5 for sampling and testing precautions. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1 liter shall be removed from each sample drum and placed in a suitable clean, dry, nonactinic bottle labeled to identify the lot and drum from which it was taken.

TABLE II. Sampling for test

Number of drums in batch or lot	Number of sample drums
1	1
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
Over 7,000	13

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled drum. Sample containers and the packaging thereof shall be examined for the following defects:

AQL 1.0 percent defective

(a) Contents per container not as specified
(b) Container damaged or leaking

AQL 2.5 percent defective

(a) Container not as specified
(b) Container closure not as specified
(c) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. See 6.5 for sampling and testing precautions. Reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.2.4.1 Assay as isopropylamine.

(a) Apparatus. Use a gas chromatograph with isothermal or temperature programming. Recommended chromatographic conditions for an F & M Model 700 chromatograph are shown in table III. These conditions may be modified, as necessary, if other chromatographs are used.

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TABLE III. Chromatographic conditions

Characteristic	Condition
Detector	Thermal conductivity
Detector current, milliamps	150
Attenuation	4X
Carrier gas	Helium
Carrier gas flow rate, mL per min	25
Block temperature	270°C
Injection port temperature	130°C
Column temperature	90°C
Column material	Stainless steel
Column dimensions	6 ft by 1/8 inch OD
Column support	80 to 100 mesh "Gas Chrom R"
Column coating	28 percent "Pennwalt 223" and 4 percent potassium hydroxide

(b) Determination of response factor. Prepare a standard solution by mixing approximately equal parts by weight of isopropylamine of known purity and diisopropylamine of known purity. Inject 0.05 microliter of the standard solution into the column. Using the conditions in table III, the approximate retention times will be 113 seconds for isopropylamine and 354 seconds for diisopropylamine. Measure the peak areas produced by the isopropylamine and the diisopropylamine. Calculate the reponse factor for isopropylamine as follows:

$$\text{Response factor} = \frac{ABC}{DEF}$$

where: A = Area of diisopropylamine peak,
B = Weight of isopropylamine injected into the column in milligrams,
C = Percent purity of the isopropylamine used to prepare the standard solution,
D = Weight of diisopropylamine injected into the column in milligrams,
E = Percent purity of diisopropylamine used to prepare the standard solution, and
F = Area of isopropylamine peak.

(c) Procedure. Prepare a test solution by mixing approximately equal parts by weight of specimen and diisopropylamine of known purity. Inject 0.05 microliter of the test solution into the column. Measure the peak areas produced by the specimen and the diisopropylamine. Calculate the percent by weight isopropylamine in the specimen as follows:

$$\text{Percent isopropylamine} = \frac{GHJK}{LM}$$

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where: G = Response factor calculated in (b),
H = Weight of diisopropylamine injected into the column in milligrams,
J = Percent purity of the diisopropylamine used to prepare the test solution,
K = Area of specimen peak,
L = Area of diisopropylamine peak, and
M = Weight of specimen injected into the column in milligrams.

4.2.4.2 Water content. Determine percent by weight water in accordance with ASTM E 203.

4.2.4.3 Specific gravity. Determine specific gravity at 20°/20°C in accordance with ASTM D 891.

4.2.4.4 Percent by volume distilled. Determine the percent by volume distilled of a 100-milliliter specimen between 30.5° and 34.5°C at 760 millimeters of mercury barometric pressure in accordance with ASTM D 1078.

5. PACKAGING

5.1 Packing, level C. Fifty-five US gallons of isopropylamine shall be packed in a 55-gallon nominal capacity steel drum which complies with Department of Transportation (DOT) regulations and Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules as applicable to the mode of transportation. The drum used shall assure safe delivery to destination without leakage or alteration of contents. The drum shall be closed as specified in the applicable rules and regulations. The drum shall be acceptable for shipment at the most favorable rate of the applicable carrier rules provided that all requirements specified have been met.

5.2 Marking. Each drum shall be marked in accordance with DOT regulations and MIL-STD-129. In addition, each container shall be durably and legibly marked with the following precautionary markings:

Hazards:

Flammable Liquid
Toxic by Inhalation
Irritant, Eye and Skin

WARNING! FLAMMABLE
HARMFUL IF INHALED
CAUSES IRRITATION

Keep away from heat, sparks and open flame.
Keep container closed.
Use with adequate ventilation.
Flash back along vapor trail may occur.
Avoid breathing vapor.

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Avoid contact with eyes, skin and clothing.
Fire extinguishing agents: Dry chemical, alcohol foam, or carbon dioxide.

FIRST AID: If inhaled, remove to fresh air.
If not breathing, give artificial respiration.
If breathing difficult, give oxygen.
In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician. Flush skin with water.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of OPA.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material (isopropylamine, flammable liquid, toxic by inhalation, irritant, eye and skin) which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

6.6 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-I-1100 dated 17 July 1972.

Custodian:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-51498(EA)	ISOPROPYLAMINE, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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